Fiscal Year:	FY 2013	**Task Last Updated:**	FY 06/25/2013
PI Name:	Kiss, John Z Ph.D.		
Project Title:	Novel Explorations Into the Interactions Between Light and Gravity Sensing in Plants		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Cellular and molecular biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Plant Biology		
Space Biology Cross-Element Discipline:	(1) Developmental Biology		
Space Biology Special Category:	(1) Bioregenerative Life Support		
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Zip Code:	27402-6170	**Congressional District:**	13
Comments:	NOTE: PI moved to Florida Institute of Technology from University of North Carolina-Greensboro in May 2024 (Ed., 2/6/25). NOTE: PI moved to University of North Carolina-Greensboro from University of Mississippi in July 2016 (Ed., 6/13/17) NOTE: PI moved to U Miss from Miami (Ohio) University in September 2012 (Ed., 8/10/2012)		
Project Type:	Flight	**Solicitation / Funding Source:**	2009 Space Life Sciences-Flight NNH09ZTT002N
Start Date:	09/11/2012	**End Date:**	09/10/2015
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:	1	**No. of Master' Degrees:**	1
No. of Master's Candidates:	2	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	2	**Monitoring Center:**	NASA ARC
Contact Monitor:	Smith, Jeffrey	**Contact Phone:**	650-604-0880
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Flight Program:	ISS		
Flight Assignment:	ISS		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Edelmann, Richard (Miami University)		
Grant/Contract No.:	NNX12AO65G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	[ED. NOTE 9/28/2012: Continuation of "Novel Explorations Into the Interactions Between Light and Gravity Sensing in Plants"--grant # NNX10AM86G, per PI move from Miami-Ohio to University of Mississippi. See that project for previous reports] The major goals of this project are to determine how gravity and light responses influence each other in plants and to better understand the cellular signaling mechanisms involved in plant tropisms. This proposed project builds on our previous spaceflight experiments on the ISS with the European Modular Cultivation System (EMCS) using the model plant Arabidopsis. In this new proposed project, we plan to confirm and extend our discovery during the recent ISS experiment of a novel red-light-based positive phototropic response in plant shoots. Our hypothesis is that positive red-light-sensing, which was known in older plant lineages, is masked by normal 1-g conditions in more recently evolved lineages. We also propose to investigate fundamental interactions among red and blue light signaling pathways and the gravity sensing mechanisms. The ISS is the only laboratory facility where it is possible to reliably and simultaneously vary light and gravity stimuli for the proposed experiments. These experiments can be accomplished with telemetric science and therefore fit with the extremely limited down mass capabilities of current spaceflight opportunities. The EMCS is an automated facility and can be operated largely with telemetry. Experimental containers will launched via Space-X, and during an experimental run, images of seedling growth responses can be downlinked to Earth. A significant advantage of this proposed project is that we will use existing flight hardware that has been flown successfully and allowed for growth of Arabidopsis seedlings in the EMCS. This project is also relevant to the recent emphasis at NASA to study plant growth and development at fractional g-levels such as those found on the moon and Mars. Improved knowledge of the basic mechanistic processes that will be the focus of this project is vital to develop ways to use plants in extraterrestrial bioregenerative life support systems.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	We plan to study the effects of gravity and light on basic cellular mechanisms in plants, so this project is relevant to the emphasis on using Arabidopsis thaliana as a model system in developmental biology. Since plants will be a necessary part of bioregenerative life support needed to send humans to Mars and beyond, the knowledge obtained from our space flight experiments will be critical for developing ways to effectively use plants in the bioregenerative life support systems. These life support systems also can be used in ground-based Earth applications such as the design of large-scale greenhouses to support crop plants and agriculture. In addition, improved knowledge of light and gravity systems in plants will, in the long term, allow us to develop better crop plants for use on Earth.
Task Progress:	Seedling Growth-1 project on the International Space Station (ISS) builds on our previous spaceflight and ground-based experiments by using the model plant Arabidopsis thaliana to study the interactions between phototropism and gravitropism in plant development. Seedling Growth-1 was successfully launched on the Space X-2 mission to the ISS in March 2013. Four runs of the experiment in the European Modular Cultivation System (EMCS) were performed on the ISS during the period from March to May 2013. The data from these experiments included video downlinks of the seedlings to analyze growth, development, and tropistic curvature at various light and gravity treatments (made possible by the variable speed centrifuge on the EMCS). Samples were frozen in the -80 C MELFI freezer and will be returned to Earth in the GLACIER freezer on a future Space X mission. There were numerous operational challenges and successes of Seedling Growth-1. In addition, several improvements in the experiments compared (to the previous TROPI experiments) included infra-red imaging of the seedlings and a novel strategy to mitigate light leaks between the rotors on the EMCS. Preliminary results on blue-light-based and red-light-based phototropism at a range of gravity levels from microgravity to fractional gravity to the 1-g control confirm the red-light-based phototropism in hypocotyls of seedlings. However, the magnitude of this novel response was reduced compared to our earlier spaceflight studies with TROPI2. Due to our interest in reduced gravity as on the Moon and Mars, this project is relevant to the long-term exploration agenda proposed by international space agencies.
Bibliography Type:	Description: (Last Updated: 11/25/2025)
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